



Geometry

Sample Test Materials

The purpose of these sample test materials is to orient teachers and students to the types of paper-based B.E.S.T. Geometry questions. By using these materials, students will become familiar with the types of items and response formats they may see on a paper-based test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test. The sample test materials are not intended to guide classroom instruction.

B.E.S.T. Geometry EOC Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches
1 yard = 3 feet
1 mile = 5,280 feet
1 mile = 1,760 yards

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces
1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters
1 meter = 1000 millimeters
1 kilometer = 1000 meters
1 liter = 1000 milliliters

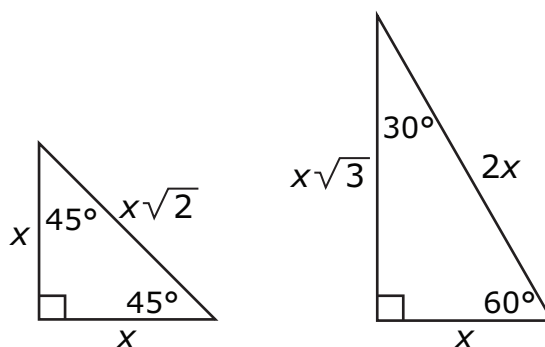
1 gram = 1000 milligrams
1 kilogram = 1000 grams

Time Conversions

1 minute = 60 seconds
1 hour = 60 minutes
1 day = 24 hours
1 year = 365 days
1 year = 52 weeks

Distance Formula	Midpoint Formula	Slope Formula
$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	$(x_M, y_M) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$	$m = \frac{y_2 - y_1}{x_2 - x_1}$

Special Right Triangles



B.E.S.T. Geometry EOC Mathematics Reference Sheet

Formulas

Parallelogram	$A = bh$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
Circle	$C = 2\pi r$ or $C = \pi d$ $A = \pi r^2$
Regular Polygon	$A = \frac{1}{2}Pa$
Prism/Cylinder	$SA = 2B + Ph$ $V = Bh$
Cone	$SA = B + \pi r h_s$ or $SA = B + \pi r l$ $V = \frac{1}{3}Bh$
Regular Pyramid	$SA = B + \frac{1}{2}Ph_s$ or $SA = B + \frac{1}{2}Pl$ $V = \frac{1}{3}Bh$
Sphere	$SA = 4\pi r^2$ $V = \frac{4}{3}\pi r^3$

Key	
P = perimeter	A = area
a = apothem	C = circumference
h = height	SA = surface area
r = radius	V = volume
h_s = slant height	
l = slant height	
b = base	
d = diameter	
B = area of base	

DO NOT REMOVE THIS SHEET

Trigonometric Ratios		
$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$	$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$	$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

Use the space in this Test and Response Book to do your work. Then, completely fill in the bubble beside the answer you choose. For some items, filling in more than one bubble may be required, so read each item carefully. If you change your answer, be sure to erase completely.

Some items will ask you to write a response in a shaded box or boxes. See the sample item below.

Sample Item:

An expression is shown.

$$(22.4)(2.65)$$

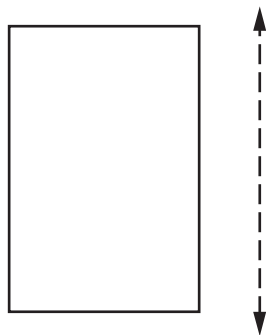
What is the value of the expression?

Write your response in the shaded box below.

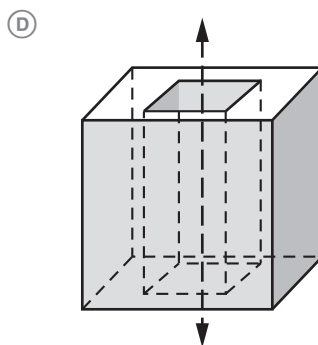
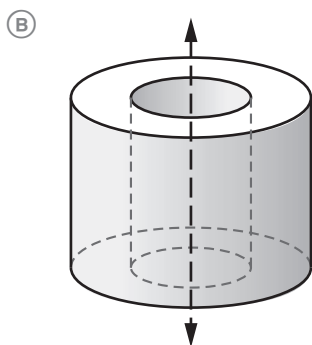
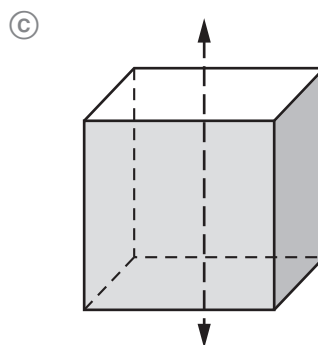
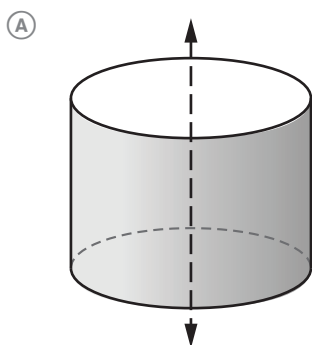
	Write your response in the box.
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Some items may have more than one box, so read each item carefully. Your answers for the items with response boxes may contain whole numbers, fractions, decimals, or negative numbers.

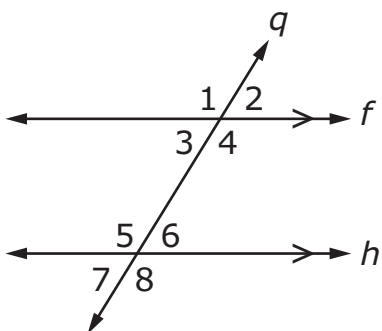
1. A rectangle and a vertical line are shown.



Which object is generated by rotating the rectangle about the vertical line?



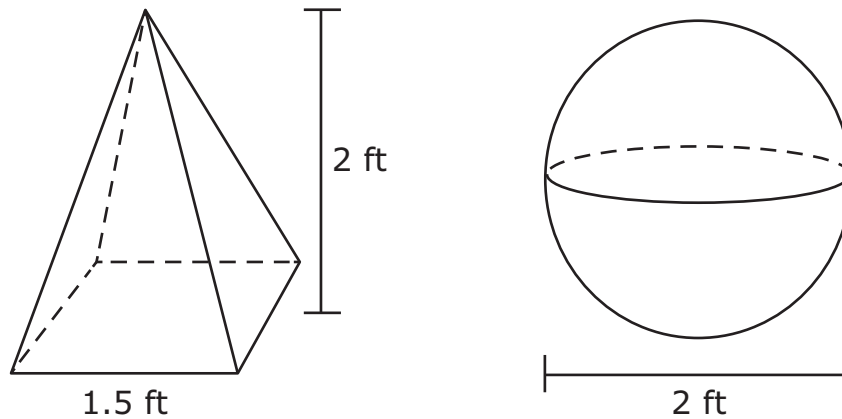
2. Lines f and h are parallel and intersected by line q , as shown.



Select statements and a reason to complete the proof showing $\angle 3 \cong \angle 6$. For each blank, fill in the bubble **before** the statement or word that is correct.

Statement	Reason
1. $f \parallel h$	1. Given
2. _____ [(A) $\angle 1 \cong \angle 6$ (B) $\angle 2 \cong \angle 6$ (C) $\angle 5 \cong \angle 6$]	2. Corresponding angles of two parallel lines cut by a transversal are congruent.
3. _____ [(A) $\angle 1 \cong \angle 3$ (B) $\angle 2 \cong \angle 3$ (C) $\angle 5 \cong \angle 3$]	3. Vertical angles are congruent.
4. $\angle 3 \cong \angle 6$	4. _____ [(A) Reflexive (B) Transitive (C) Symmetric] property of congruence

3. A right square pyramid and a sphere are made of marble. The dimensions of each figure, in feet (ft), are shown.

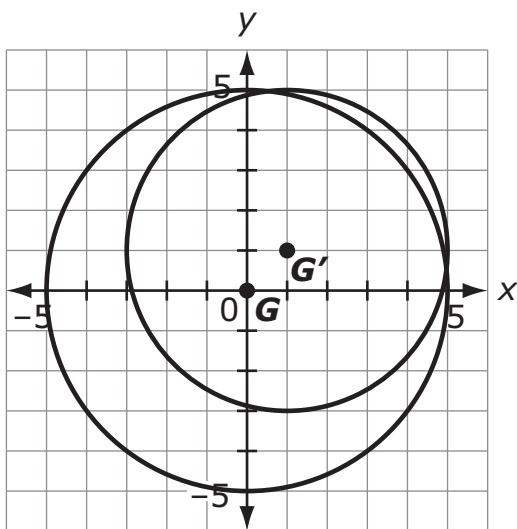


The density of marble is 160 pounds per cubic foot.

What is the difference, in pounds, between the weight of the sphere and the weight of the pyramid? Round your answer to the nearest hundredth.

Write your response in the shaded box below.

4. The preimage and image of a circle are shown.



The center of circle G is at $(0, 0)$, and the center of circle G' is at $(1, 1)$. The radius of circle G is 5 units, and the radius of circle G' is 4 units.

The sequence of transformations applied to circle G to create circle G' is shown.

- $(x, y) \rightarrow (cx, cy)$
- $(x, y) \rightarrow (x + d, y + d)$

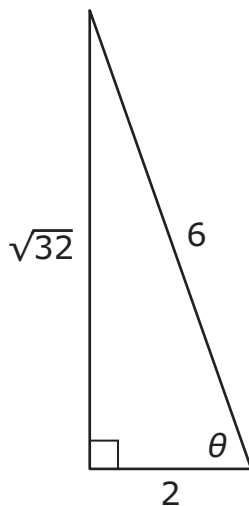
What are the values of c and d ?

Write your responses in the shaded boxes below.

$c =$

$d =$

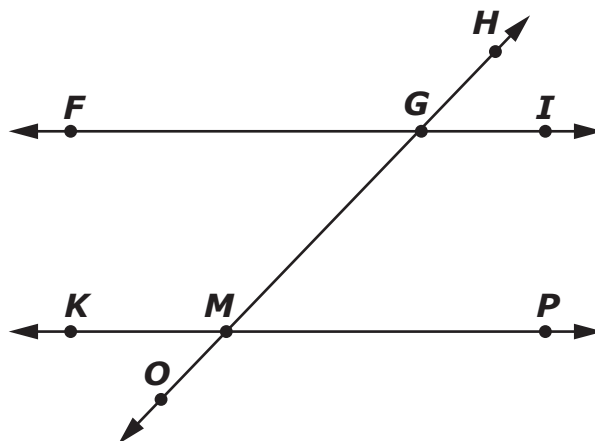
5. A figure is shown.



Select values to complete the trigonometric ratio. For each part of the ratio, fill in the bubble **before** the number that is correct.

$$\tan(\theta) = \frac{[\textcircled{A} 2 \textcircled{B} \sqrt{32} \textcircled{C} 6]}{[\textcircled{A} 2 \textcircled{B} \sqrt{32} \textcircled{C} 6]}$$

6. A diagram is shown.



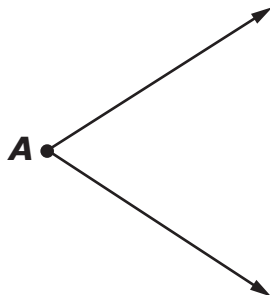
Kori uses the diagram to analyze the given statement.

“When a transversal intersects parallel lines, two angles are supplementary if and only if they are adjacent angles.”

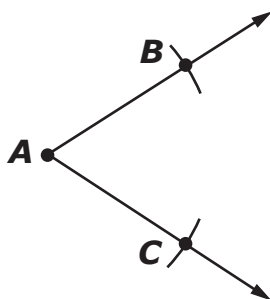
Fill in the bubble **before** one angle from each column to show a pair of angles that represents a counterexample.

Angle 1	Angle 2
☐ A $\angle FGH$	☐ I $\angle FGH$
☐ B $\angle FGM$	☐ J $\angle FGM$
☐ C $\angle PMG$	☐ K $\angle PMG$
☐ D $\angle PMO$	☐ L $\angle PMO$
☐ E $\angle HGI$	☐ M $\angle HGI$
☐ F $\angle IGM$	☐ N $\angle IGM$
☐ G $\angle KMG$	☐ O $\angle KMG$
☐ H $\angle KMO$	☐ P $\angle KMO$

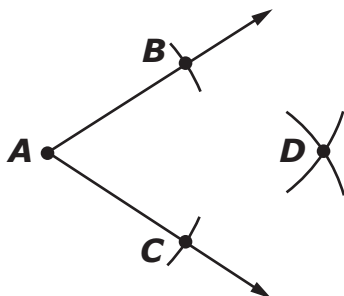
7. Angle A and two steps of a construction are shown.



Step 1: Place the compass at point A and draw arcs that intersect both rays. Label the intersections as points B and C , as shown.



Step 2: Using the same compass width from Step 1, draw intersecting arcs in the interior of the angle from points B and C . Label the intersection as point D , as shown.



This question has **two** parts.

Part A

Select phrases to create a true statement. For each blank, fill in the bubble **before** the phrase that is correct.

Constructing $\overline{AD}$ will create two congruent angles, _____

[☐ (A) $\angle ABC$ and $\angle ACB$ ☐ (B) $\angle ABD$ and $\angle ACD$ ☐ (C) $\angle BAD$ and $\angle CAD$], because $\overline{AD}$ is _____ [☐ (A) a median ☐ (B) an incenter ☐ (C) an angle bisector].

Part B

Select all the lengths that are equivalent to AB .

- ☐ (A) AC
- ☐ (B) AD
- ☐ (C) BC
- ☐ (D) BD
- ☐ (E) CD

8. Angelo transforms rectangle $PQRS$ into rectangle $JKLM$. The corresponding sides of rectangles $PQRS$ and $JKLM$ are **not** congruent.

Select all the transformations that Angelo could have used.

- Ⓐ $(x, y) \rightarrow (y, x)$
- Ⓑ $(x, y) \rightarrow (3x, 3y)$
- Ⓒ $(x, y) \rightarrow (-y, x)$
- Ⓓ $(x, y) \rightarrow (1.5x, 1.5y)$
- Ⓔ $(x, y) \rightarrow (x - 2, y + 3)$

B.E.S.T. Mathematics Sample Items

9. Point $P(-1, 4)$ lies on line segment QR with endpoints $Q(1, 7)$ and $R(-9, -8)$.

Fill in bubbles to match each ratio of segment lengths to the correct numerical ratio.

	1:4	1:5	4:5
$QP:PR$	(A)	(B)	(C)
$QP:QR$	(D)	(E)	(F)



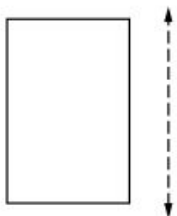
Geometry

Sample Items

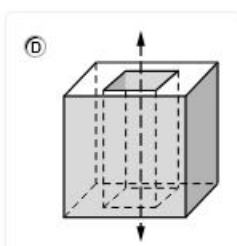
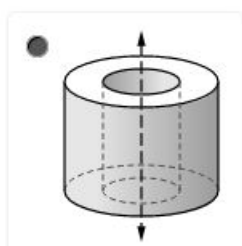
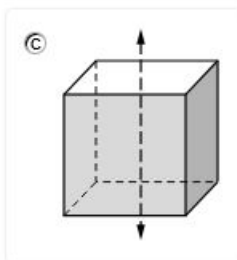
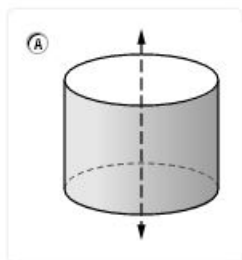
Answer Key

The B.E.S.T. Geometry Sample Items Answer Key provides the correct response(s) for each item on the sample test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test.

1. A rectangle and a vertical line are shown.



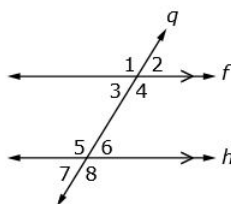
Which object is generated by rotating the rectangle about the vertical line?



Option B: **This answer is correct.** The student determined that rotating the rectangle about a vertical line would generate a cylinder with a hole through the middle of it when the rectangle does not touch the vertical line.

B.E.S.T. Mathematics Sample Items Answer Key

2. Lines f and h are parallel and intersected by line q , as shown.

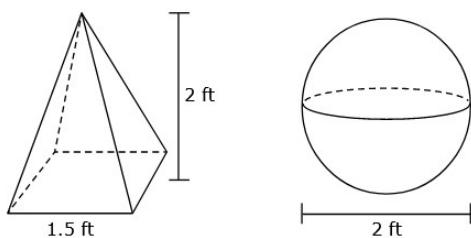


Select statements and a reason to complete the proof showing $\angle 3 \cong \angle 6$.

Statement	Reason
1. $f \parallel h$	1. Given
2. $\angle 2 \cong \angle 6$	2. Corresponding angles of two parallel lines cut by a transversal are congruent.
3. $\angle 2 \cong \angle 3$	3. Vertical angles are congruent.
4. $\angle 3 \cong \angle 6$	4. Transitive property of congruence

B.E.S.T. Mathematics Sample Items Answer Key

3. A right square pyramid and a sphere are made of marble. The dimensions of each figure, in feet (ft), are shown.



The density of marble is 160 pounds per cubic foot.

What is the difference, in pounds, between the weight of the sphere and the weight of the pyramid? Round your answer to the nearest hundredth.

430.21

← → ↶ ↷ ✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Other correct responses: any value between -430.48 and -429.86 or 429.86 and 430.48 , inclusive

B.E.S.T. Mathematics Sample Items Answer Key

4. A scientist discovers a spherical object in the solar system with a mass of 1.1×10^{25} grams and a volume of 2.9×10^{24} cubic centimeters. He uses the table shown to classify the planet based on its density.

Planet	Density
Gas	Between 0.3 and 2.1 grams per cubic centimeter
Rocky	Between 3.6 and 13.4 grams per cubic centimeter

Select a word and enter a value to classify the object. Round to the nearest tenth if necessary.

The object can be classified as a planet because its density is grams per cubic centimeter.

Other correct responses: Any value between 3.79 and 3.8 is acceptable.

- 5.** Jalen constructs the circumscribed circle of the triangle shown.

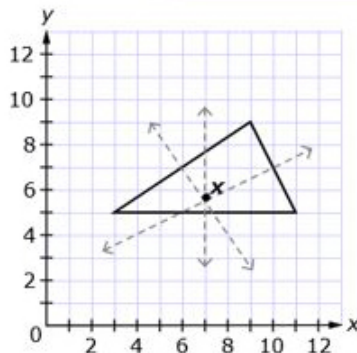
- A. Select the line type Jalen uses to construct the circle.
- B. Drag point x to the location of the center of the circumscribed circle.

A. Line Types

Altitudes
Medians

Angle bisectors
Perpendicular bisectors

B.



B.E.S.T. Mathematics Sample Items Answer Key

6. The vertices of one of the diagonals of a square are located at $(-3, 4)$ and $(2, -1)$.

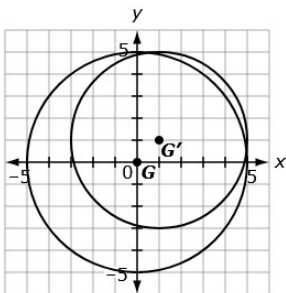
Enter values and select a phrase to describe the location of the other vertices of the square.

The other vertices of the square must be located at $(2, 4)$ and $(-3, -1)$ because those points make the diagonals congruent and perpendicular. ⚙

←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
	0			
.	-	$\frac{\Box}{\Box}$		

Note: The ordered pairs may be switched.

7. The preimage and image of a circle are shown.



The center of circle G is at $(0, 0)$, and the center of circle G' is at $(1, 1)$. The radius of circle G is 5 units, and the radius of circle G' is 4 units.

The sequence of transformations applied to circle G to create circle G' is shown.

- $(x, y) \rightarrow (cx, cy)$
- $(x, y) \rightarrow (x + d, y + d)$

What are the values of c and d ?

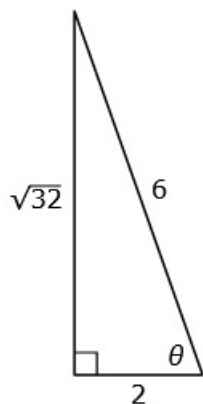
$c =$

$d =$

← → ↶ ↷ ✕		
1	2	3
4	5	6
7	8	9
	0	
.	-	□

Other correct responses: any equivalent values

8. A figure is shown.



Select values to complete the trigonometric ratio.

$$\tan(\theta) = \frac{\sqrt{32}}{2}$$

B.E.S.T. Mathematics Sample Items Answer Key

9. The equation of a circle is shown.

$$x^2 + 8x + y^2 - 12y = -27$$

Enter values to show the coordinates of the center and the length of the radius of the circle.

Center: (,)

Radius: units

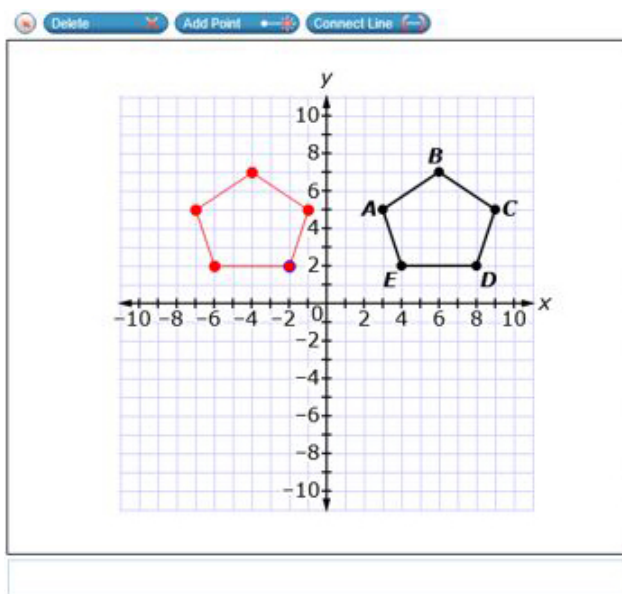
Other correct responses: any equivalent values

B.E.S.T. Mathematics Sample Items Answer Key

10. Sal has a pentagon $ABCDE$.

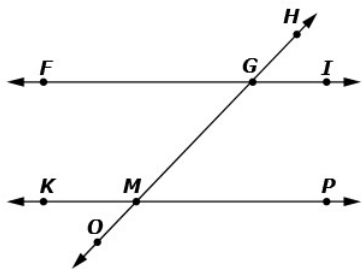
- He translates $ABCDE$ 2 units left.
- Then, he reflects the resulting image over the y -axis to create pentagon $LMNOP$.

Use the Connect Line tool to draw $LMNOP$.



Note: The student may draw both the final image and the pentagon after the translation and get credit.

11. A diagram is shown.



Kori uses the diagram to analyze the given statement.

“When a transversal intersects parallel lines, two angles are supplementary if and only if they are adjacent angles.”

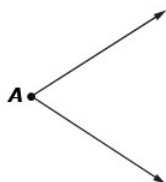
Select one angle from each column to show a pair of angles that represents a counterexample.

Angle 1	Angle 2
$\angle FGH$	$\angle FGH$
$\angle FGM$	$\angle FGM$
$\angle PMG$	$\angle PMG$
$\angle PMO$	$\angle PMO$
$\angle HGI$	$\angle HGI$
$\angle IGM$	$\angle IGM$
$\angle KMG$	$\angle KMG$
$\angle KMO$	$\angle KMO$

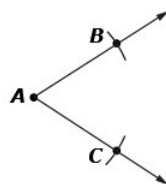
Note: Any two supplementary angles that are not adjacent angles are acceptable.

B.E.S.T. Mathematics Sample Items Answer Key

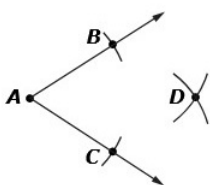
12. Angle A and two steps of a construction are shown.



Step 1: Place the compass at point A and draw arcs that intersect both rays. Label the intersections as points B and C , as shown.



Step 2: Using the same compass width from Step 1, draw intersecting arcs in the interior of the angle from points B and C . Label the intersection as point D , as shown.



This question has **two** parts.

Part A

Select phrases to create a true statement.

Constructing $\overline{AD}$ will create two congruent angles, $\angle BAD$ and $\angle CAD$, because $\overline{AD}$ is an angle bisector.

Part B

Select all the lengths that are equivalent to AB .

☒ AC

☐ AD

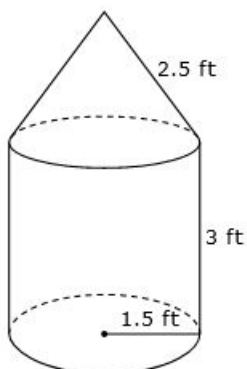
☐ BC

☒ BD

☒ CD

B.E.S.T. Mathematics Sample Items Answer Key

- 13.** A figure composed of a right cone and a right cylinder with dimensions in feet (ft) is shown.



What is the exact surface area, in square feet, of the figure?

15π

←
→
↶
↷
✖

1	2	3	+	-	•	÷				
4	5	6	<	≤	=	≥	>			
7	8	9	$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i
	0		sin	cos	tan	arcsin	arccos	arctan		
.	-	$\frac{\square}{\square}$								

Other correct responses: any equivalent value

B.E.S.T. Mathematics Sample Items Answer Key

- 14.** Angelo transforms rectangle $PQRS$ into rectangle $JKLM$. The corresponding sides of rectangles $PQRS$ and $JKLM$ are **not** congruent.

Select all the transformations that Angelo could have used.

☐ $(x, y) \rightarrow (y, x)$

☒ $(x, y) \rightarrow (3x, 3y)$

☐ $(x, y) \rightarrow (-y, x)$

☒ $(x, y) \rightarrow (1.5x, 1.5y)$

☐ $(x, y) \rightarrow (x - 2, y + 3)$

Option B: **This answer is correct.** The student correctly identified that the transformation represents a dilation of 3, which does not preserve distance.

Option D: **This answer is correct.** The student correctly identified that the transformation represents a dilation of 1.5, which does not preserve distance.

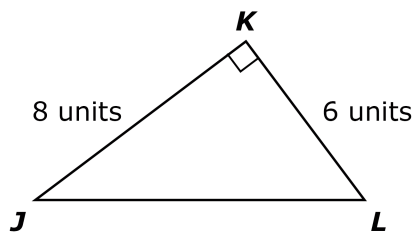
B.E.S.T. Mathematics Sample Items Answer Key

- 15.** Point $P(-1, 4)$ lies on line segment QR with endpoints $Q(1, 7)$ and $R(-9, -8)$.

Match each ratio of segment lengths to the correct numerical ratio.

	1:4	1:5	4:5
$QP:PR$	☒	☐	☐
$QP:QR$	☐	☒	☐

- 16.** Right triangle JKL is shown, with dimensions in units.



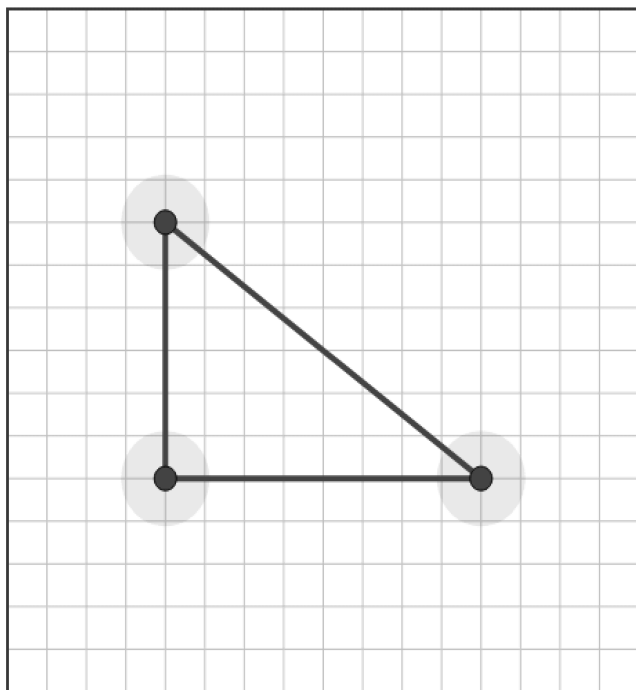
Twyla creates a congruent triangle based only on the given dimensions.

B.E.S.T. Mathematics Sample Items Answer Key

This question has **two** parts.

Part A

Click on the grid to draw Twyla's triangle. Each square on the grid has a side length of 1 unit.



Part B

Using the given dimensions, what theorem can Twyla use to prove the triangle she creates is congruent to triangle JKL ?

☐ Ⓐ Angle-Side-Angle

☒ Ⓑ Side-Angle-Side

☐ Ⓒ Hypotenuse-Leg

☐ Ⓓ Angle-Angle

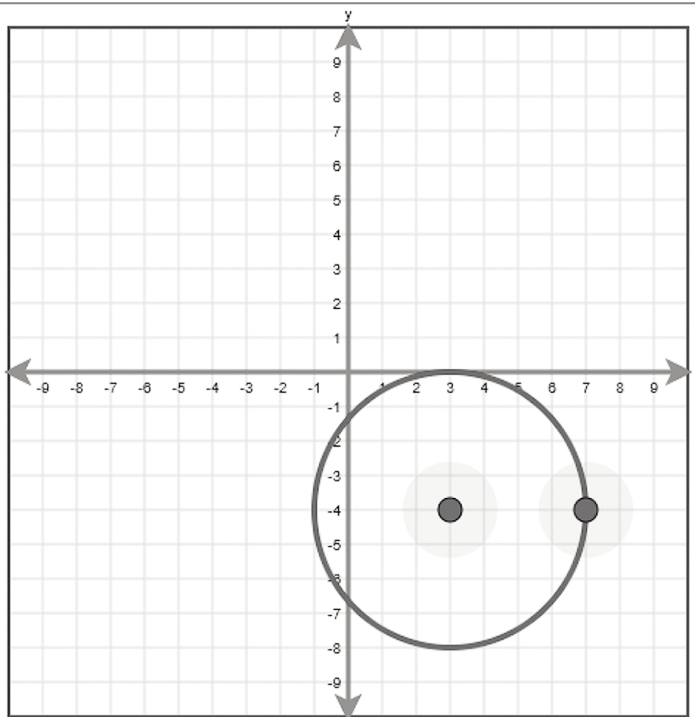
Option B: This answer is correct. The student identified the theorem that can be used to prove congruence with the given information.

Other correct responses: for Part A, any right triangle with legs 8 units long and 6 units long

- 17.** The equation of a circle is given.

$$(x - 3)^2 + (y + 4)^2 = 16$$

Plot the center and a point on the circle to graph the circle.



Other correct responses: any circle with a center at $(3, -4)$ and a radius of 4